



Basic Principles of Medicine 1
Module: Musculoskeletal System | DLA

Clinical Anatomy of the Shoulder

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Recommended Reading

- Drake, Vogl, Mitchell. Grays Anatomy for Students. 4th ed. Elsevier Churchill Livingstone.
- Printed Textbook: Chapter 7
- Online Textbook: Chapter 7: [click here](#)
 - Regional Anatomy:
 - Shoulder
- Pay special attention to the **green (printed text)** / **blue (ebook)** “*In the Clinic*” boxes for the clinical correlations for this topic

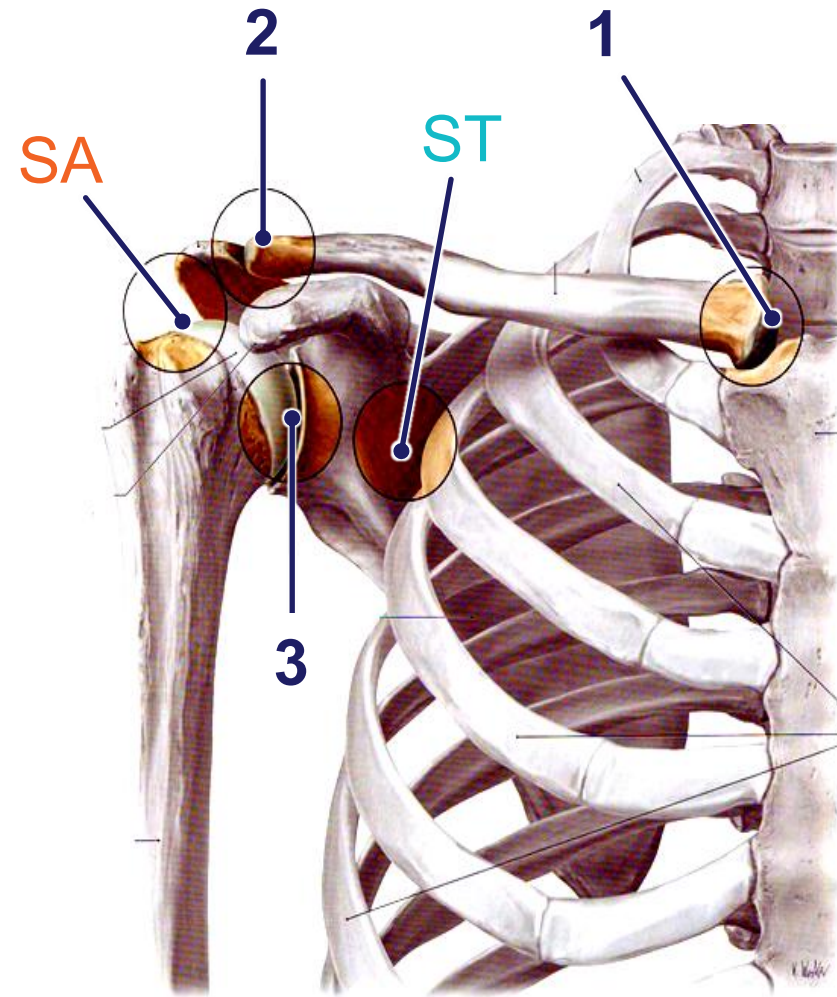


Objectives

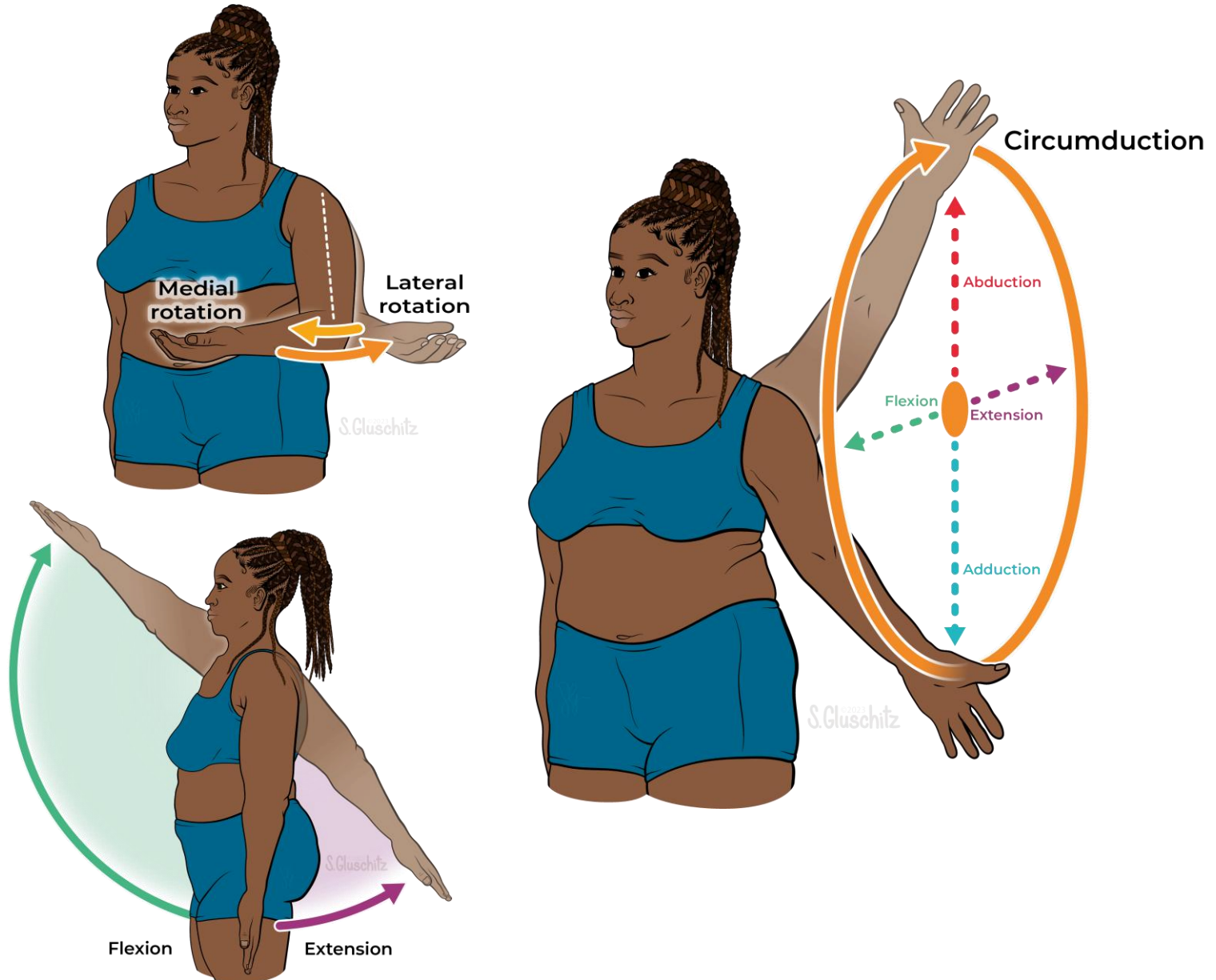
- SOM.MKII.BPM1.1.MSK.1.ANAT.1136** Describe the movements that occur at the shoulder joint
- SOM.MKII.BPM1.1.MSK.1.ANAT.1137** Identify, name and orient the bones forming the shoulder girdle and the shoulder joint.
- SOM.MKII.BPM1.1.MSK.1.ANAT.1138** Describe the 3 joints that form the shoulder complex
- SOM.MKII.BPM1.1.MSK.1.ANAT.1139** Describe the location of the subacromial and subscapular bursae at the shoulder joint.
- SOM.MKII.BPM1.1.MSK.1.ANAT.1140** Describe the ligamentous injuries and clinical signs in acromioclavicular (shoulder) separation and identify this on imaging studies.
- SOM.MKII.BPM1.1.MSK.1.ANAT.1141** Describe the signs and symptoms of clavicular fracture, and how the fragments of the fracture would be displaced
- SOM.MKII.BPM1.1.MSK.1.ANAT.1142** List the major elements that contribute to the stability of the glenohumeral joint.
- SOM.MKII.BPM1.1.MSK.1.ANAT.1143** Describe the rotator cuff muscles and their function in maintaining the stability of the glenohumeral joint.
- SOM.MKII.BPM1.1.MSK.1.ANAT.1144** Explain why the glenohumeral joint can be easily dislocated.
- SOM.MKII.BPM1.1.MSK.1.ANAT.1145** Describe shoulder dislocation injuries, particularly antero - inferior dislocations, and explain the nerve injuries that might accompany shoulder dislocations.
- SOM.MKII.BPM1.1.MSK.1.ANAT.1146** Identify antero - inferior shoulder dislocation on a radiograph.
- SOM.MKII.BPM1.1.MSK.1.ANAT.1147** Describe the anatomical and clinical signs of rotator cuff disorders due to injury of the supraspinatus muscle
- SOM.MKII.BPM1.1.MSK.1.ANAT.1148** Describe the cutaneous innervation of the shoulder.
- SOM.MKII.BPM1.1.MSK.1.ANAT.1149** List the attachments, innervations and actions of the muscles that act on the shoulder joint.

Shoulder Joint Complex

- Where the upper limb attaches to the trunk
- Bones: humerus, scapula, clavicle, thoracic cage
- 3 synovial joints
 - Shoulder girdle (scapula and clavicle)
 - 1) Sternoclavicular
 - 2) Acromioclavicular
 - Shoulder joint proper
 - 3) Glenohumeral
- Scapulothoracic joint
- Subacromial space
 - Common site of impingement

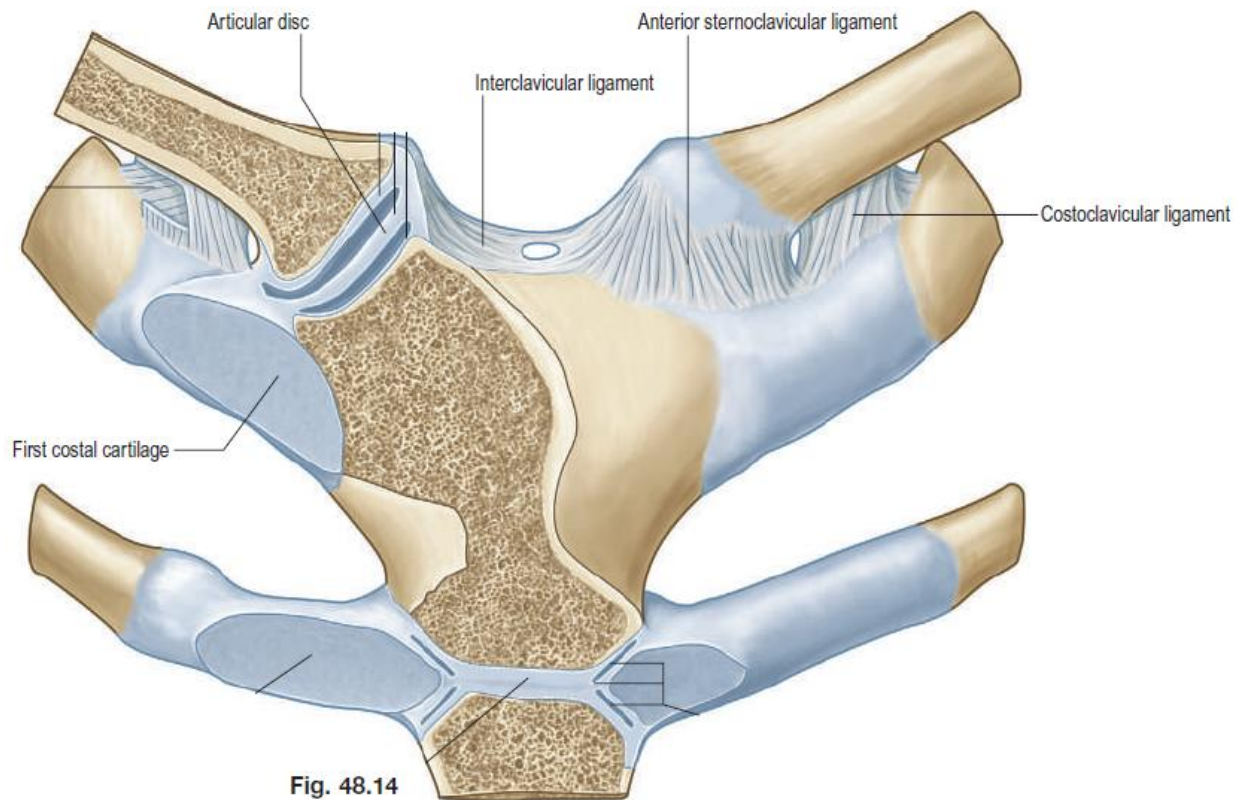


Movements at the Shoulder Joint



Sternoclavicular joint

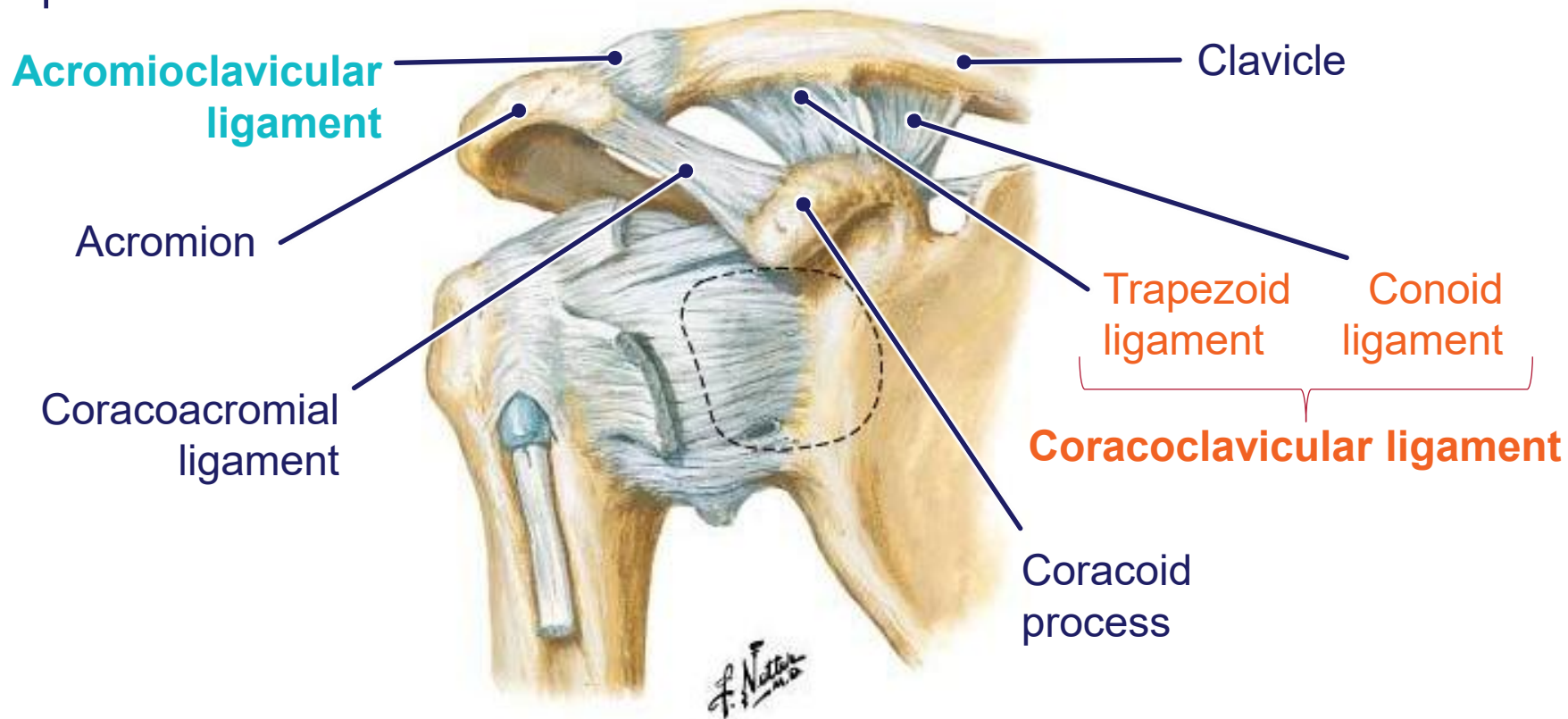
- Only articulation of upper limb to axial skeleton
- Stable joint



Acromioclavicular joint

Reinforced by:

- 1. **Acromioclavicular ligament**: strengthens the acromioclavicular joint capsule superiorly
- 2. **Coracoclavicular ligament**: anchors the clavicle to the coracoid process



Clinical Correlate: Fractured Clavicle

A 16-year-old cheerleader jumped and landed on the tip of her shoulder during practice.

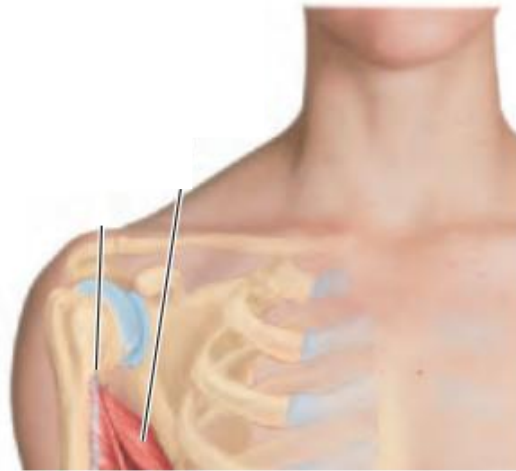
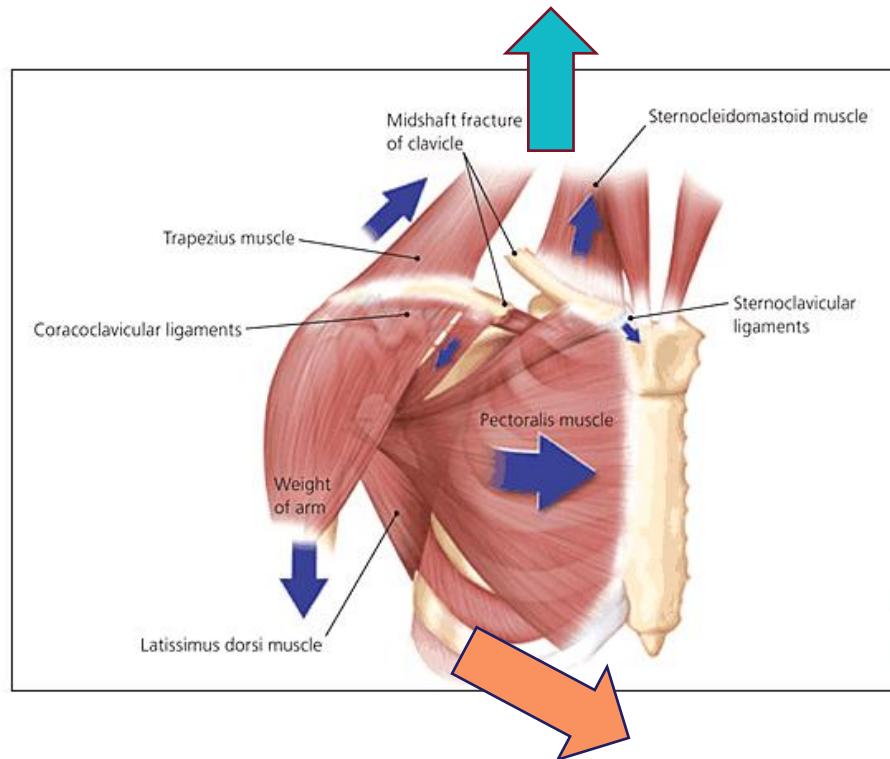


FIGURE 6.26.



Clinical Correlate: Fractured Clavicle

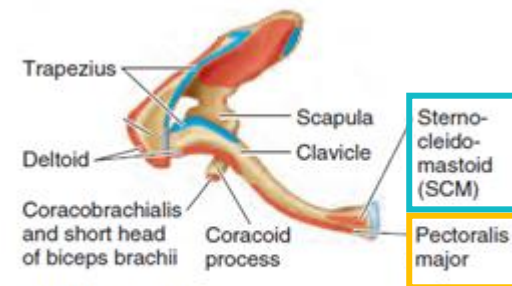


Midshaft fracture of clavicle

- Shoulder depressed
- Arm medially rotated and adducted

Possible associated intrathoracic injuries:

- Pneumothorax
- Damage to subclavian vein and artery

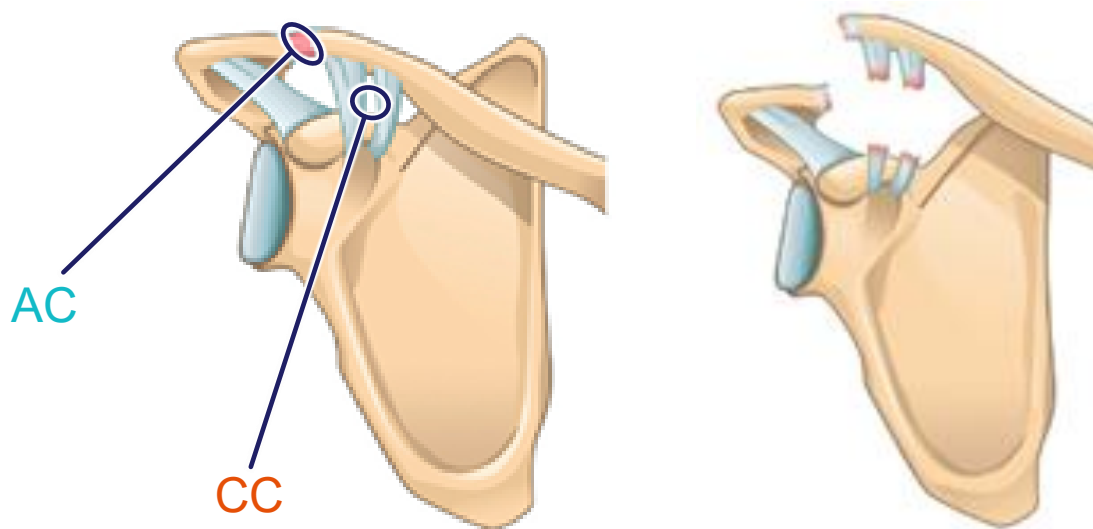


(B) Superior view

FIGURE 6.23.

Clinical Correlate: Shoulder Separation

- A 26-year-old man fell directly on his right shoulder with the arm in an outstretched and overhead position. Pain and swelling were immediate, and were associated with a “step deformity”

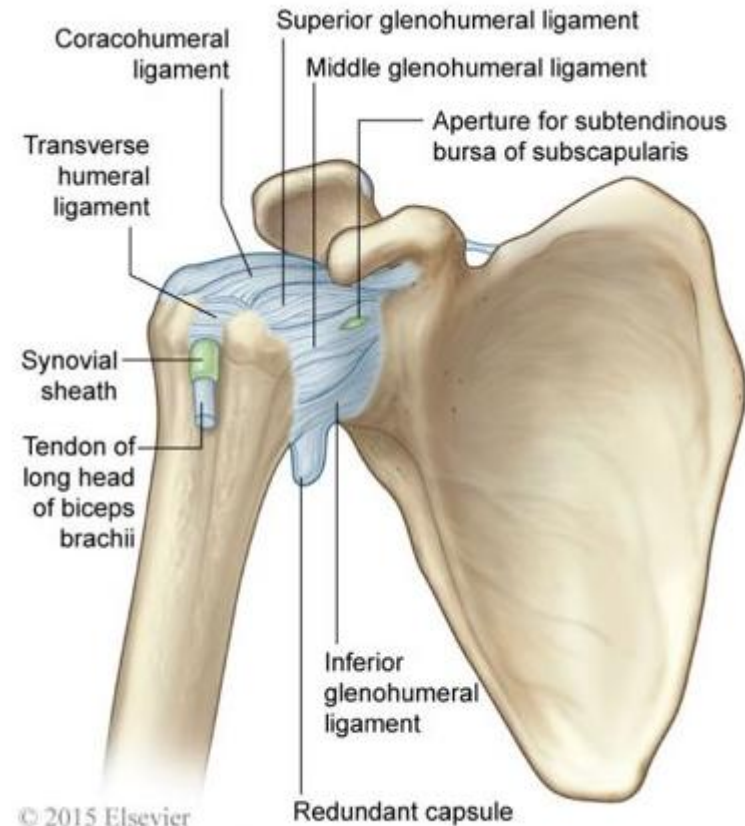


Flores et al., Radiographics 40 (5):1355-1382, 2020

- Varying degrees of severity
- Tearing of the **acromioclavicular (AC) joint and ligament**
 - with or without the **coracoclavicular (CC) ligament**
- Restricted range of shoulder movement

Glenohumeral joint overview

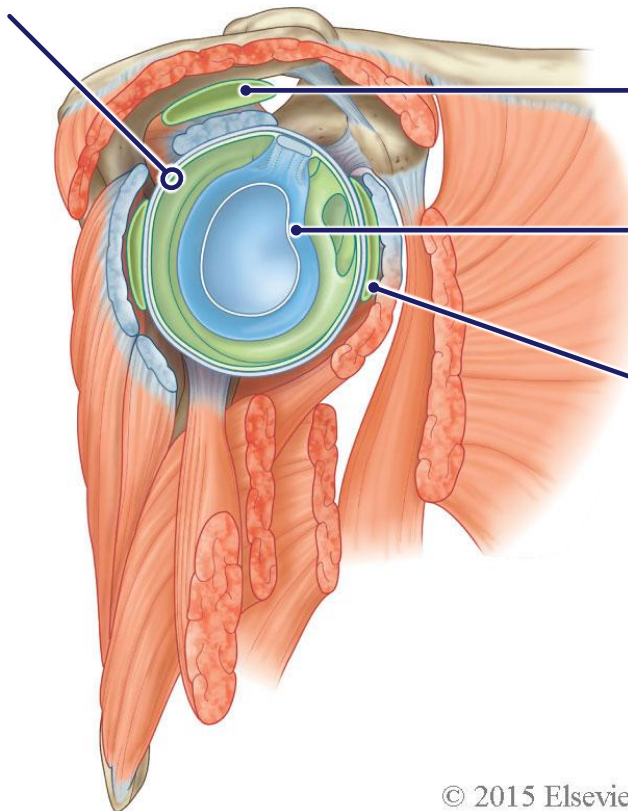
- Enveloped by a glenohumeral capsule
- Attaches to the anatomical neck of the **humerus** and **neck of scapula**
- Reinforced by glenohumeral ligaments and tends of **SITS muscles**
- Wide range of motion
 - Inverse relationship between range of motion and stability



Glenohumeral joint

- Deepened by the **fibrocartilaginous collar (glenoid labrum)**
- **Subdeltoid (subacromial) bursa** allows for easy movement of the supraspinatus tendon under the coracoacromial arch of the deltoid over the greater tubercle

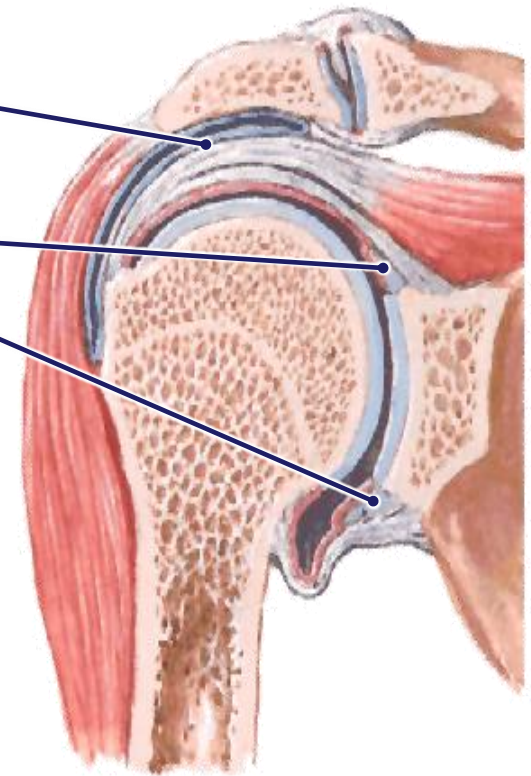
Synovial membrane



Subdeltoid bursa

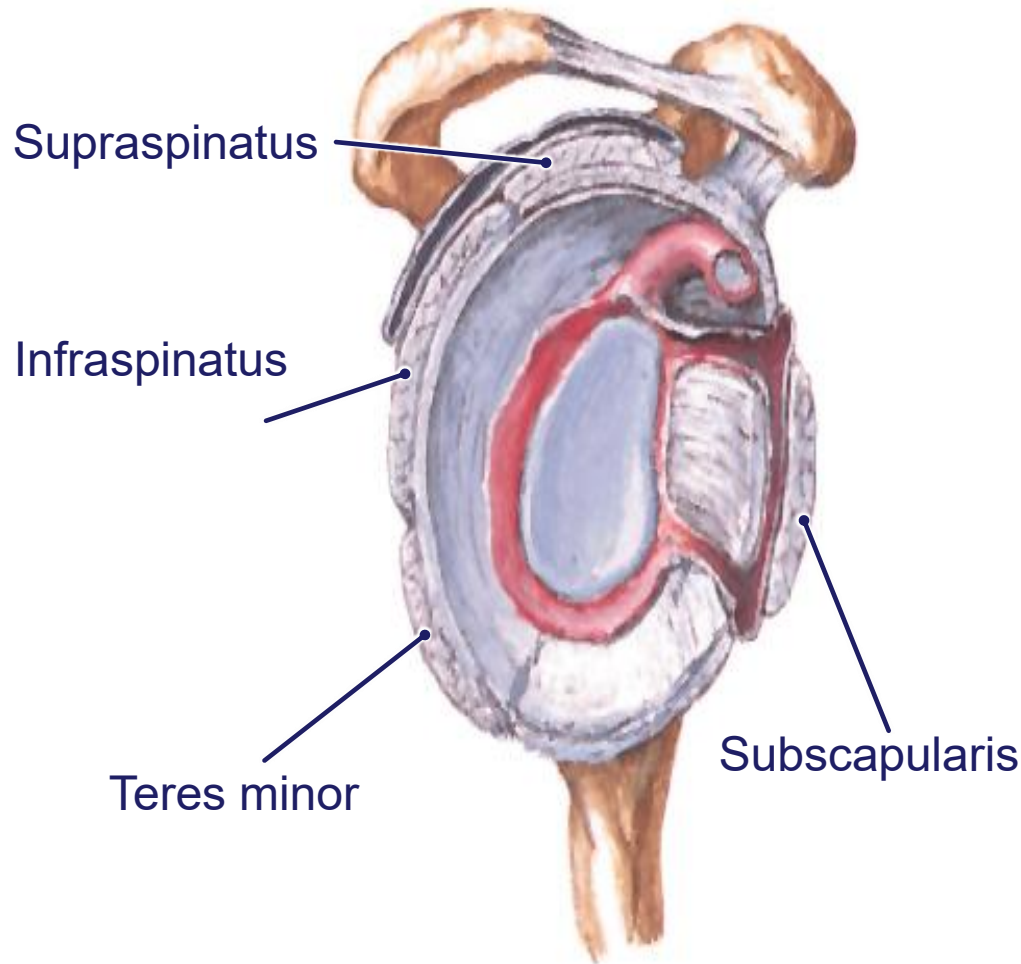
Glenoid labrum

Subtendinous bursa of subscapularis

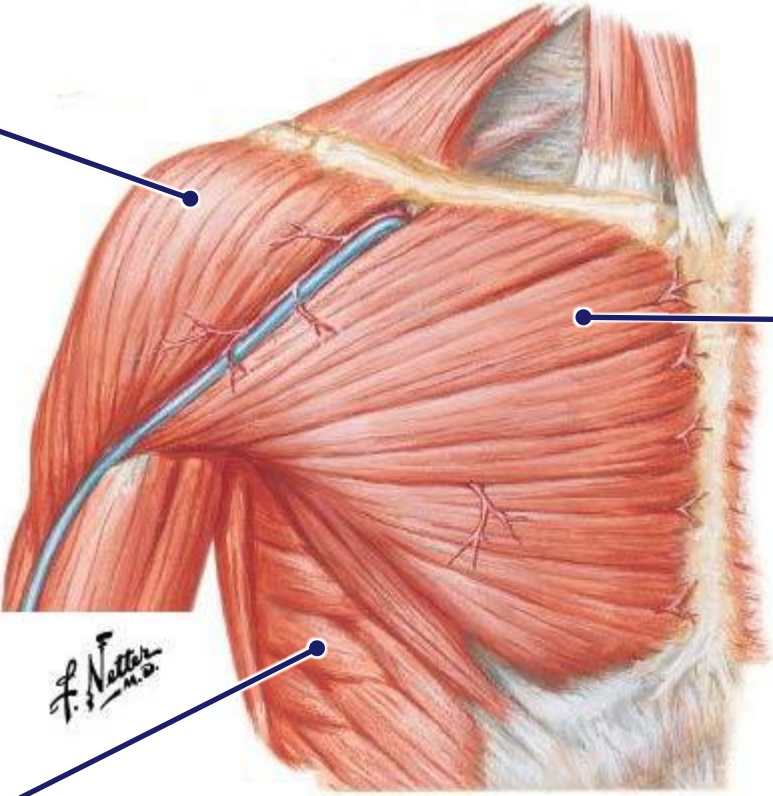


Joint Capsule Reinforcement

The rotator cuff muscles play a major role in reinforcement of the joint capsule.



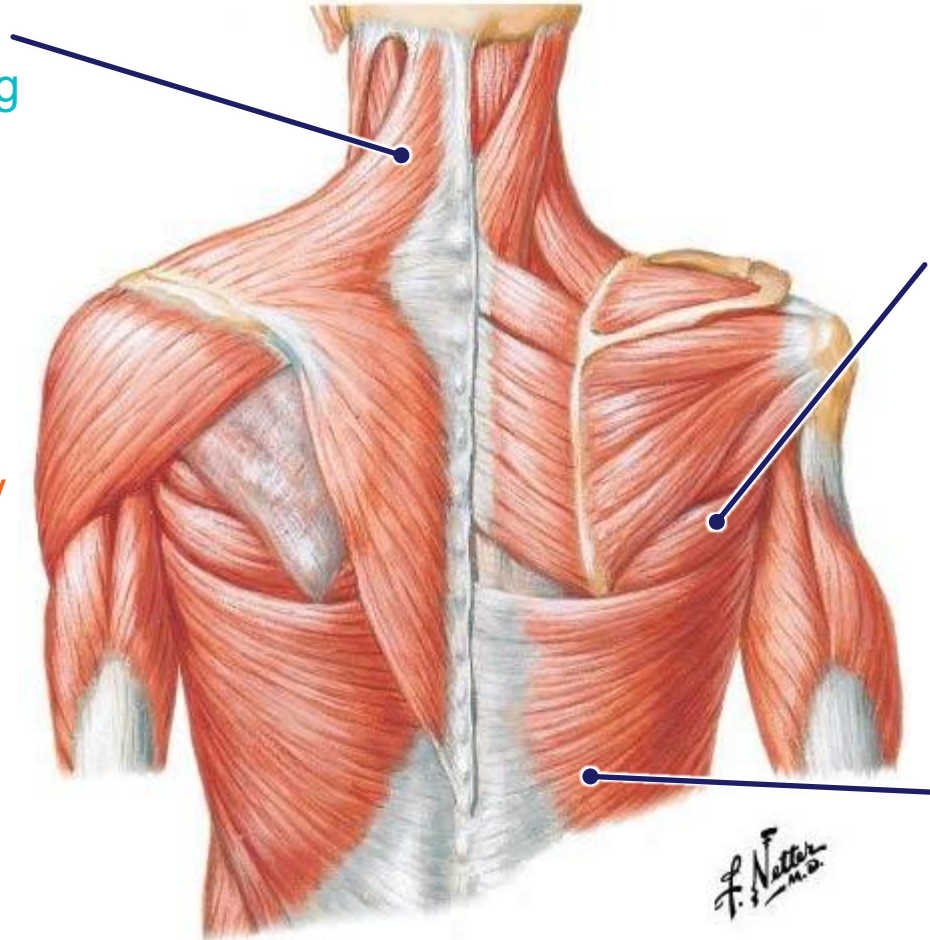
Muscles acting on the shoulder joint

- 
- Deltoid**
 - **Abducts** arm 15 - 90°
 - **Axillary nerve**
 - Pectoralis major**
 - **Flexion, adduction, and medial rotation** of the arm
 - **Medial and lateral pectoral nerves**
 - Serratus anterior**
 - **Rotates scapula upward for abduction** of arm above 90°
 - **Protracts the scapula**
 - **Long thoracic nerve (C5,6,7)**

Muscles acting on the shoulder joint

Trapezius

- Assists in rotating the scapula upwards during **abduction** of the arm above 90°
- Elevation of the scapula
- **Spinal accessory nerve**



Teres major

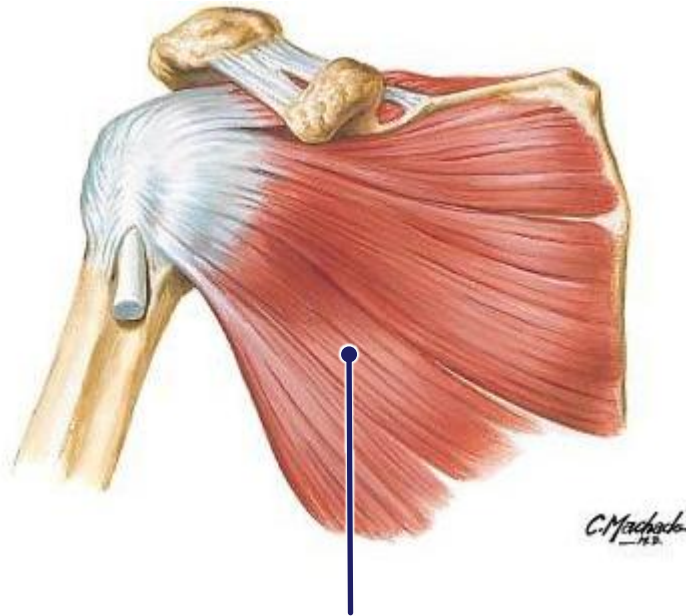
- **Extends, adducts, and medially rotates the arm**
- **Lower subscapular nerve**

Latissimus dorsi

- **Extends, adducts, and medially rotates the arm**
- **Thoracodorsal nerve**

Muscles acting on the shoulder joint:

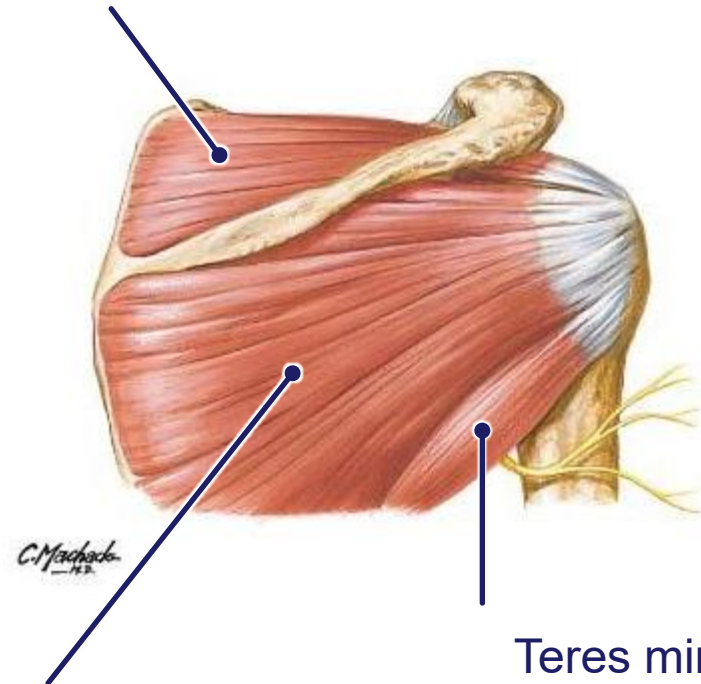
Rotator cuff muscles



Subscapularis

- Medial rotation of the arm
- Upper and lower subscapular nerve

- Supraspinatus
- Abduction of the arm to 15°
 - Suprascapular nerve



Infraspinatus

- Lateral rotation of the arm
- Suprascapular nerve

- Teres minor
- Lateral rotation of the arm
 - Axillary nerve

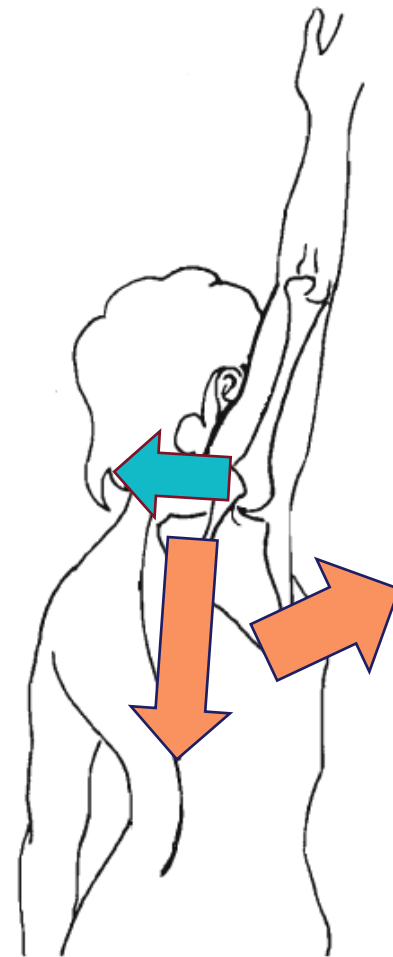
Arm Abduction - recap



Supraspinatus



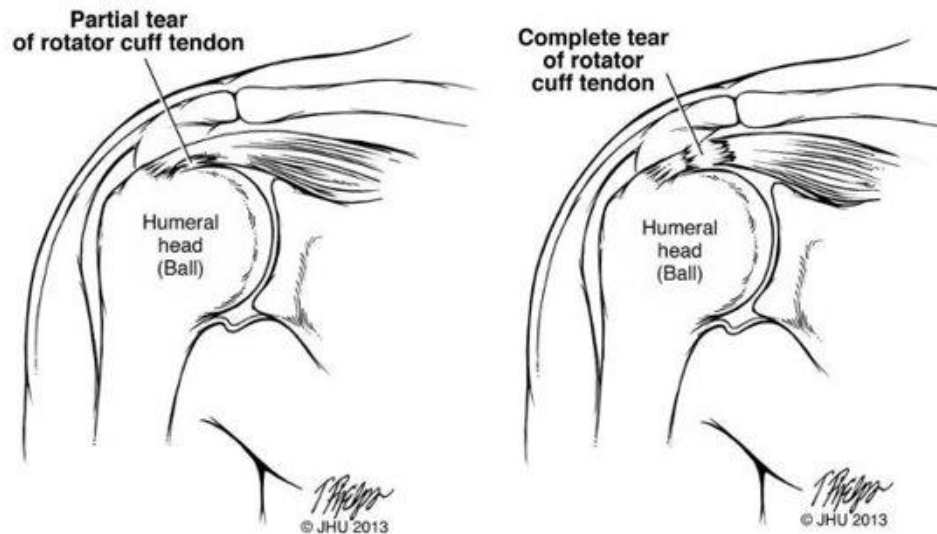
Deltoid



Trapezius
Serratus anterior

Clinical Correlate: Rotator Cuff Injuries

- Most commonly involves ***supraspinatus***
 - Susceptible to impingement and degenerative changes (calcification) due to repetitive trauma
 - Can be impinged by calcified **coracoacromial ligament**, swelling of the supraspinatus muscle, or bony subacromial spurs
- Produces severe pain on **abduction**
- Leads to partial or full thickness tears
- Can result in the overlying **subacromial bursa** becoming inflamed



Clinical Correlate: Shoulder Dislocation

- An 18-year-old football player was recovering a fumbled ball when he was struck in the shoulder by two other players.
- On examination there was **prominence of the acromion with “squaring” of the shoulder contour.**

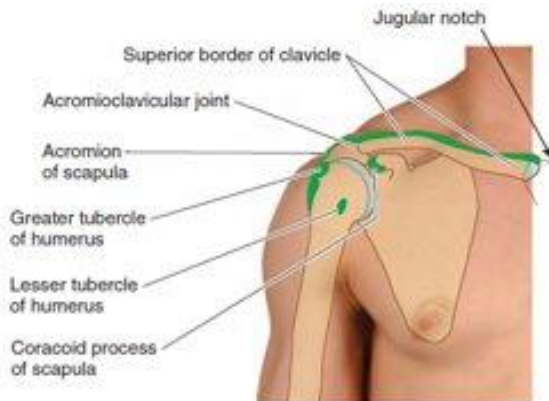
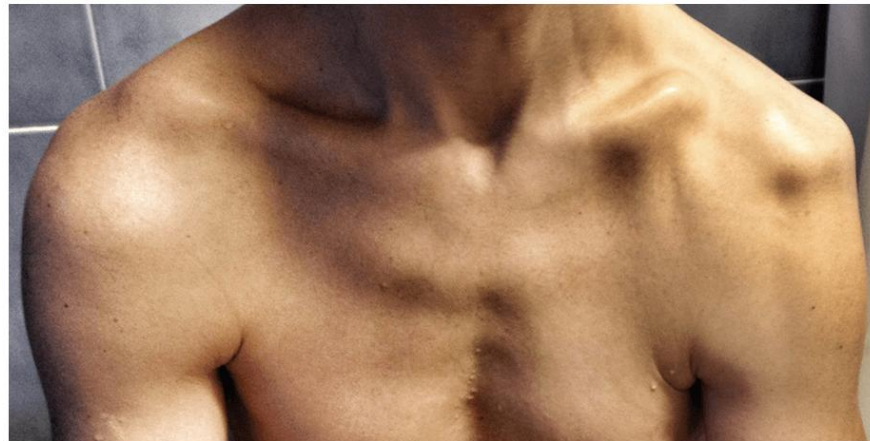
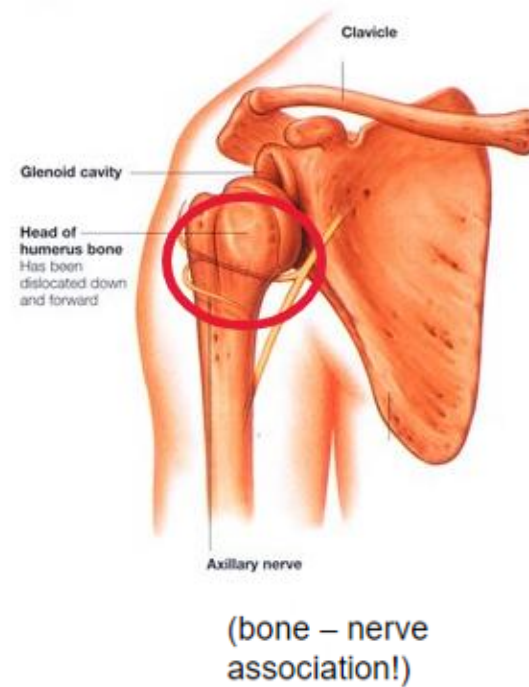
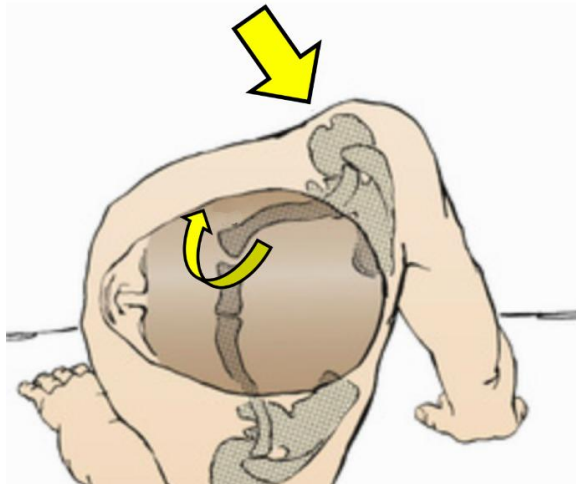


FIGURE 6.10



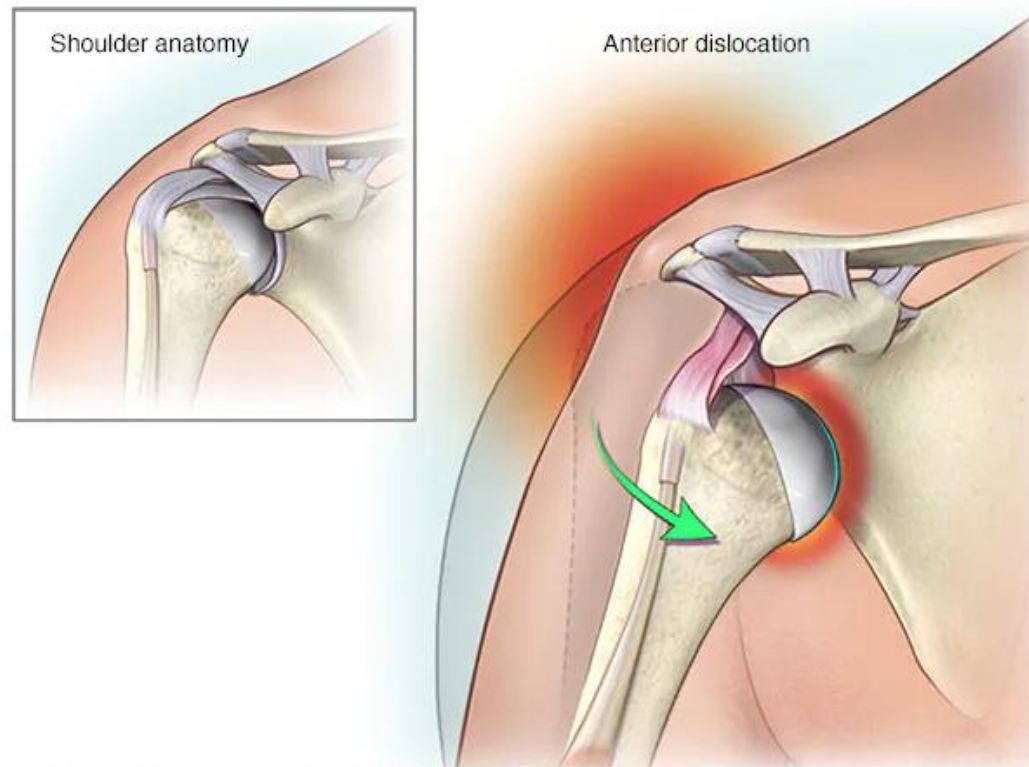
Clinical Correlate: Shoulder Dislocation

- An 18-year-old football player was recovering a fumbled ball when he was struck in the shoulder by two other players.
- The patient was in severe pain, and kept his shoulder immobilized in **slight abduction with external rotation**.



Clinical Correlate: Shoulder Dislocation

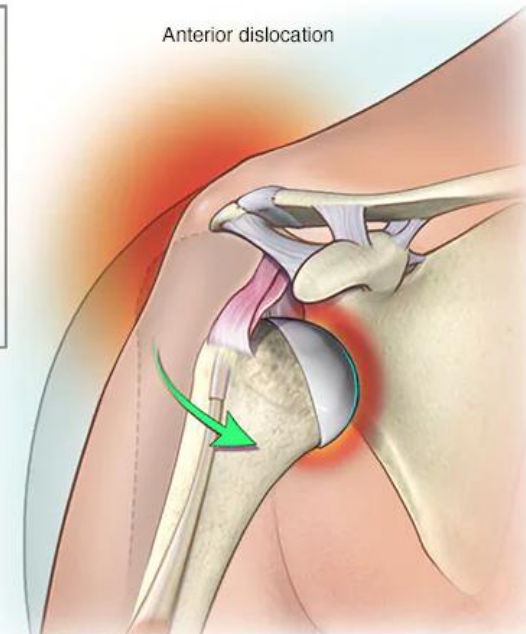
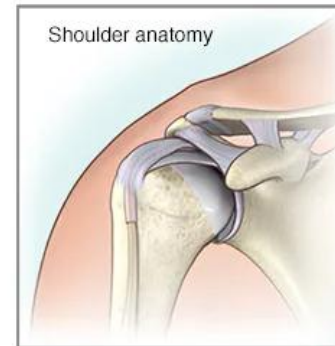
- Why is the dislocation antero-inferior?
- What neurovascular structures can be damaged in this patient?
- What sensory and motor deficits may be present in this patient?



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Clinical Correlate: Shoulder Dislocation

- Why is the dislocation antero-inferior?
 - How much support is present?
- What neurovascular structures can be damaged in this patient?
 - **Axillary nerve** and **posterior humeral circumflex artery**
- What sensory and motor deficits may be present in this patient?
 - Decreased ability to **laterally rotate** the arm
 - Loss of **abduction** from 15 to 90°
 - Sensory loss in **upper lateral** part of arm (regimental patch)



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Innervation & Blood Supply of the Shoulder Joint

- **Innervation**

- Follows **Hilton's law**: nerves providing motor innervation to the muscles acting on a joint will also carry sensation from that joint
- Branches from the posterior cord of the brachial plexus, terminal branches of the brachial plexus and from the suprascapular, axillary and lateral pectoral nerves

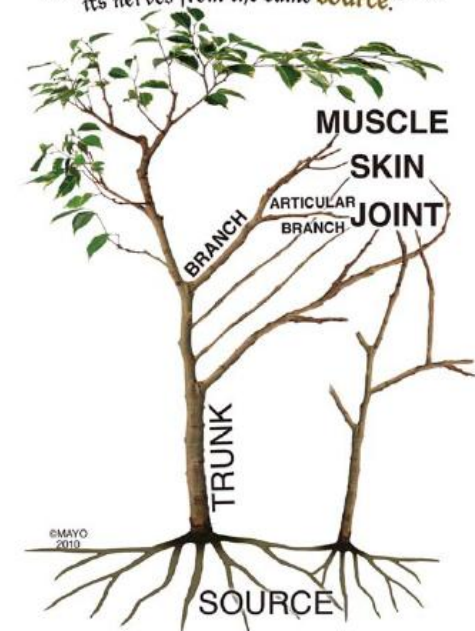
- **Blood supply**

- Branches of the anterior and posterior circumflex humeral and suprascapular arteries

- **Venous drainage**

- Paired veins of the same name, accompany the arteries that supply the shoulder joint

"The same **trunks** of nerves whose **branches** supply the groups of muscles moving a joint furnish also a distribution of nerves to the skin over the insertions of the same muscles; and, what at this moment more especially merits our attention, the interior of the joint receives its nerves from the same **source**."



Hebert-Blouin et al., Clin Anat 27:548-555, 2014

Cutaneous Innervation: Shoulder

Supraclavicular
nerves

- C3 and C4 from
cervical plexus

Superior lateral brachial
cutaneous nerve

- Axillary nerve



FIN

